

Quiz (Carolina Reaper)

- Q1.** A shipping company charges 2 per package plus 0.50 per mile. If x represents the number of miles, what is the algebraic expression for the total cost?
- (A) $0.50x + 2$
 (B) $2x + 0.50$
 (C) $2x - 0.50$
 (D) $0.50 - 2x$
 (E) $2 - 0.50x$
- Q2.** Evaluate the expression $3 \cdot 2 + 11 - 5$ using the correct order of operations.
- (A) 10
 (B) 15
 (C) 20
 (D) 6
 (E) 4
- Q3.** A traffic flow model uses the equation $t = \frac{d}{r}$, where t is time, d is distance, and r is rate. If $d = 250$ miles and $r = 50$ miles per hour, what is the value of t ?
- (A) 5 hours
 (B) 3 hours
 (C) 2 hours
 (D) 8 hours
 (E) 10 hours
- Q4.** Simplify the algebraic expression $2x + 5x - 3x + 2$ by combining like terms.
- (A) $4x + 2$
 (B) $6x - 2$
 (C) $4x - 2$
 (D) $6x + 2$
 (E) $8x - 1$
- Q5.** A delivery company uses the equation $c = 2w + 5$ to calculate the total cost, where c is the total cost and w is the weight of the package in pounds. If the weight of the package is 10 pounds, what is the total cost?
- (A) 25
 (B) 20
 (C) 15
 (D) 10
 (E) 5
- Q6.** Evaluate the expression $(3 + 2) \cdot (8 - 3)$ using the correct order of operations.
- (A) 20
 (B) 30
 (C) 25
 (D) 15
 (E) 5
- Q7.** The equation $d = rt$ represents the distance traveled by a vehicle, where d is distance, r is rate, and t is time. If $r = 40$ miles per hour and $t = 2.5$ hours, what is the distance traveled?
- (A) 100 miles
 (B) 80 miles
 (C) 120 miles
 (D) 60 miles
 (E) 50 miles
- Q8.** Simplify the algebraic expression $x^2 + 4x - 3x - 2$ by combining like terms.
- (A) $x^2 + x - 2$
 (B) $x^2 - 3x + 2$
 (C) $x^2 + 2x - 1$
 (D) $x^2 - x - 2$
 (E) $x^2 + 3x - 2$

Open Ended Questions (FRQ)

- Q9.** A logistics company has a fleet of trucks with different capacities. The total capacity of the trucks can be represented by the equation $C = 2x^2 + 5x - 3$, where C is the total capacity and x is the number of trucks. Find the total capacity when there are 4 trucks. Show your work and explain your answer.
- Q10.** A traffic engineer is designing a new highway system. The time it takes for a car to travel from point A to point B can be represented by the equation $t = \frac{d}{r}$, where t is time, d is distance, and r is rate. If the distance between point A and point B is 120 miles and the rate is 40 miles per hour, find the time it takes for a car to travel from point A to point B. Additionally, if the rate increases by 10 miles per hour, how much time is saved?

Chef's Tips

- Tip 1: Make sure to evaluate expressions using the correct order of operations (PEMDAS).
- Tip 2: Encourage students to show their work and explain their answers.
- Tip 3: Remind students to check their units when solving problems involving distance, rate, and time.